11.2 A 0.85fJ/conversion-step 10b 200kS/s Subranging SAR ADC in 40nm CMOS

Hung-Yen Tai, Yao-Sheng Hu, Hung-Wei Chen, Hsin-Shu Chen

National Taiwan University, Taipei, Taiwan

Analog-to-digital converters (ADC) are extensively used in wireless sensor networks and healthcare electronic devices to monitor long-term signal conditions. It is essential to prolong battery life in these applications by using an energy-efficient ADC. A successive-approximation register (SAR) architecture, mostly composed of digital circuits, can achieve low power under low supply voltages [1,2]. Power consumption can be decreased by using either an energy-efficient capacitive-DAC switching method [1] or a low-power comparator with a majority voting technique [2]. In this work, a small coarse ADC resolves the MSB bits. Then, a detect-and-skip algorithm and an aligned switching technique are used to reduce the big fine DAC switching energy. The comparator power is also decreased by utilizing a low-power comparator during coarse conversion and a low-noise comparator during fine conversion. As a result, its FoM performance is as low as 0.85fJ/conversion-step, which is about 3 times better than that of the state-of-the-art work [2].

Figure 11.2.1 shows the architecture of our subranging SAR ADC. It comprises a 5b coarse SAR ADC, a 10b fine SAR ADC, and skipping control logic. The main idea is to relieve the requirements of the fine ADC by a coarse ADC whose accuracy constraint is greatly reduced by redundancy. During the sampling phase, the clock signal CLK turns on 4 bootstrap sampling switches. The input signals are sampled onto the coarse and fine capacitor arrays. During the conversion phase, the coarse ADC detects the sampled signal with the results of B1 to B5, and then the skipping control logic skips unnecessary capacitor switching. An asynchronous signal C_DONE passes the aligned switching signals to switch the MSB capacitors of the fine DAC. Next, the fine ADC continues to resolve the remaining bits of B6 to B11. The gains of coarse and fine DACs are matched by adding a C_D in the coarse DAC. An additional bit resolving [3] in the fine ADC with the redundancy range of ±16 LSB relieves the accuracy requirement on the coarse ADC. In addition, the DAC gain and comparator offset mismatches between the coarse and fine ADCs can also be compensated by the redundancy. The coarse DAC and the MSB capacitors of the fine DAC adopt a split-capacitor switching method [4] to keep the comparator common-mode voltage constant. Thus, the comparator dynamic offset due to common-mode variation does not affect the linearity. Moreover, the LSB capacitors in the fine DAC apply a monotonic switching method [3] to save switching energy. Because the DAC output common-mode variation with the LSB bits is relatively small, the dynamic offset error is negligible.

Figure 11.2.2 illustrates the detect-and-skip algorithm. The coarse ADC resolves B1 to B5. It detects the difference between the fine DAC output and the sampled signal. When the difference is small, unnecessary switching of MSB capacitors in the fine DAC can be skipped and then the DAC output would directly approach the sampled signal. Otherwise, energy is wasted by MSB capacitor switching with the conventional binary-search algorithm of SAR ADC. Since the MSB bits control large capacitors, most of energy is wasted during the resolving of the MSB bits. Our detect-and-skip algorithm saves much switching energy of the fine DAC with small switching energy of the coarse DAC. If the difference is small, B1 and B2 are 01 or 10. The MSB capacitor switching of the fine DAC can be skipped to save power. On the other hand, if the difference is large, B1 and B2 are 00 or 11. The MSB capacitor should be switched. Similarly, when B1 and B3 are 01 or 10, the MSB-1 capacitor switching can be skipped. The same principle can be applied to MSB-2 and MSB-3 capacitors as well. Figure 11.2.2 also depicts the aligned switching technique, which reduces switching energy when the difference is large. The plot of switching energy versus the ratio of C_{SW} and C_T shows that the largest switching energy occurs when half of capacitors are switched as in conventional successive switching algorithm. For example, when 3 unit capacitors (2C and 1C) have to be switched successively, the total switching energy is calculated to be $1.25CV^2$. However, if 2C and 1C are switched simultaneously by our aligned switching technique, the switching energy is as small as $0.75CV^2$. Hence, to switch the MSB capacitors of the fine DAC with the aligned switching technique is more energy-efficient than successive switching.

Figure 11.2.3 shows the switching energy versus output code. The average switching energy in the conversion phase is only $69.8CV^2$ (coarse $9.7CV^2$ and fine $60.1CV^2$), which is lower than the other switching methods. The total average switching energy is $229.7CV^2$ which includes a pre-charge energy of $159.9CV^2$ (coarse $15CV^2$ and fine $144.9CV^2$) in the sampling phase. It relaxes the settling requirement of the reference voltage by setting small portion of switching energy to the conversion phase. Also, the reference disturbance resulting from pre-charge is not an issue for the long settling time in the sampling phase. The total average switching energy of this work is larger than that of the MCS switching method [5]. But the MCS switching method uses $0.5V_{DD}$ in addition to V_{DD} and ground as reference voltages, which is hard to implement in low-voltage designs.

A 2-stage dynamic comparator [6] is used in both coarse and fine ADCs. Its noise level is the bottleneck of accuracy in low-voltage design. The accuracy requirement of the coarse comparator can be relieved from 10b to 5b with redundancy. Thus, a low-power comparator is implemented in the coarse ADC and a low-noise comparator is implemented in the fine ADC. The first-stage size of the coarse comparator is identical to that of the fine comparator to reduce the offset mismatch. The input-referred noise and comparator power are proportional to $1/\sqrt{C_L}$ and C_L, respectively, where C_L is the output capacitance loading of the first stage. Hence, C_L in the coarse comparator could be smaller than that of the fine comparator. The power-per-bit-conversion of the coarse comparator is designed to be 3 times less than that of the fine comparator. As a result, it saves 23% of power compared with the traditional SAR that uses only one 10b noise-level comparator.

Figure 11.2.7 shows the chip micrograph of the ADC fabricated in 40nm CMOS. The core circuit occupies an area of $0.0065mm^2$ ($110\mu m \times 59\mu m$). The capacitor array is constructed by metal–oxide–metal (MOM) type capacitors with unit capacitance of 1.5fF. The measured DNL and INL are shown in Fig. 11.2.4. At conversion rate F_s of 200kS/s, the DNL and INL are +0.29/-0.44 and +0.45/-0.29 LSB, respectively. The Nyquist rate FFT plot and dynamic performance versus input frequency are shown in Fig. 11.2.5. With Nyquist input frequency, the measured SNDR, SFDR, SNR, and THD are 55.63dB, 76.25dB, 55.75dB, and 71.3dB, respectively. The measured power dissipation is 84nW with a 0.45V supply. It can be broken down as follows: the DACs use 34% of the power; the comparators and the bootstraps use 30%; the digital circuits use 36%. Figure 11.2.6 shows the performance summary and comparison table of the energy-efficient ADCs in recent years.

Acknowledgements:
The authors would like to thank Taiwan National Science Council and TSMC University Shuttle Program for the support of this work. They would also like to thank HYCON technology for the help of measurement.

References:
[1] C.-Y. Liou and C.-C. Hsieh, "A 2.4-to-5.2fJ/conversion-step 10b 0.5-to-4MS/s SAR ADC with Charge-Average Switching DAC in 90nm CMOS," *ISSCC Dig. Tech. Papers*, pp. 280-281, Feb. 2013.
[2] P. Harpe, *et al.*, "A 2.2/2.7fJ/conversion-step 10/12b 40kS/s SAR ADC with Data-Driven Noise Reduction," *ISSCC Dig. Tech. Papers*, pp. 270-271, Feb. 2013.
[3] C.-C. Liu, *et al.*, "A 10b 100MS/s 1.13mW SAR ADC with Binary-Scaled Error Compensation," *ISSCC Dig. Tech. Papers*, pp. 386-387, Feb. 2010.
[4] M. Yip and A. P. Chandrakasan, "A Resolution-Reconfigurable 5-to-10b 0.4-to-1V Power Scalable SAR ADC," *ISSCC Dig. Tech. Papers*, pp. 190-191, Feb. 2011.
[5] V. Hariprasath, *et al.*, "Merged Capacitor Switching Based SAR ADC with Highest Switching Energy-Efficiency," *IET Electronics Letters*, Apr. 2010.
[6] M. van Elzakker, *et al.*, "A 1.9μW 4.4fJ/Conversion-step 10b 1MS/s Charge-Redistribution ADC," *ISSCC Dig. Tech. Papers*, pp. 244-245, Feb. 2008.

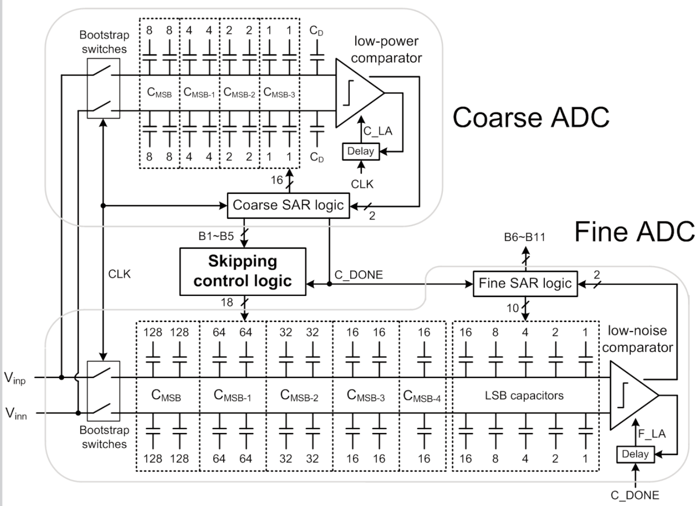


Figure 11.2.1: Subranging SAR ADC.

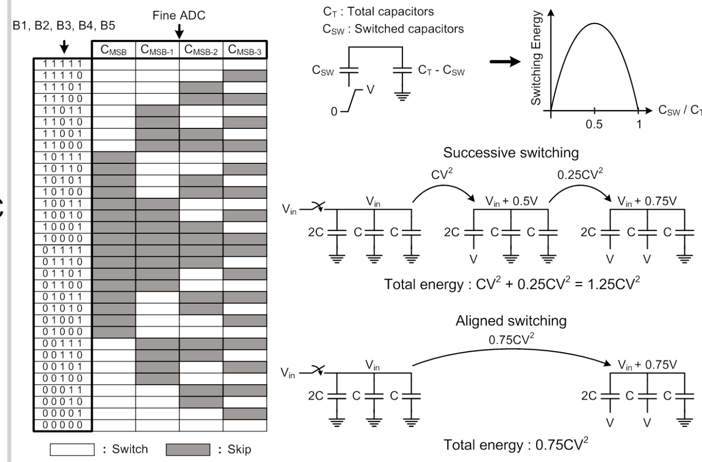
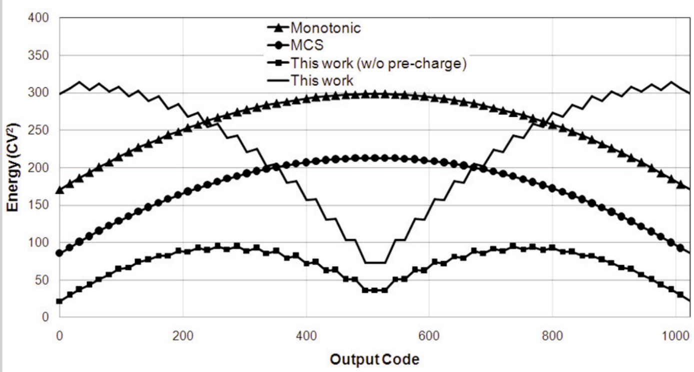


Figure 11.2.2: Detect-and-skip algorithm and aligned switching technique.



Switching method	Average energy (CV^2)
Monotonic	255.5
MCS	170.2
This work	229.7
This work (w/o pre-charge)	69.8

Figure 11.2.3: Switching energy versus output code.

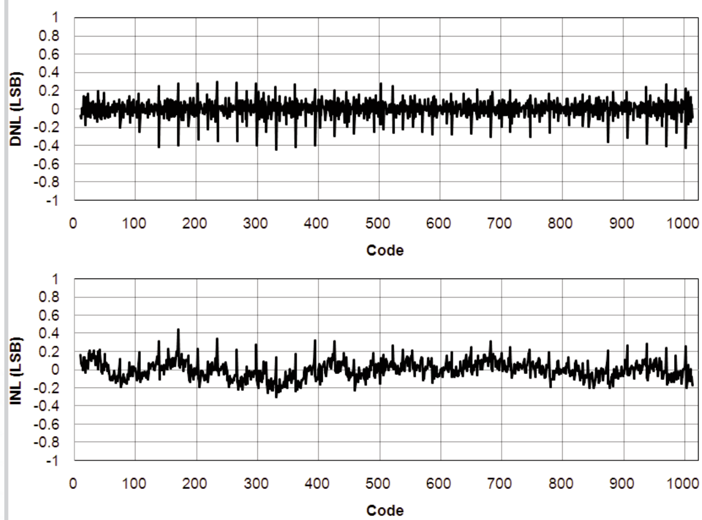


Figure 11.2.4: DNL and INL.

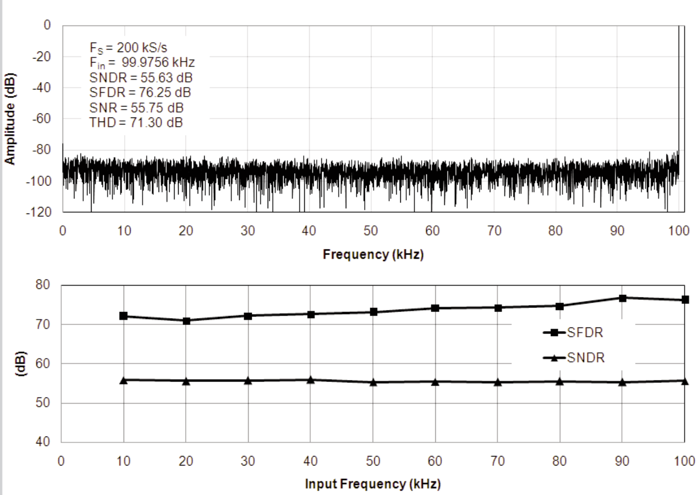


Figure 11.2.5: Nyquist rate FFT and dynamic performance versus input frequency at 200kS/s.

	[1]	[2]	[6]	This work
Technology	90nm	65nm	65nm	40nm
Supply Voltage (V)	0.4	0.6	1	0.45
Sample rate (kS/s)	500	40	1000	200
Resolution (bit)	10	12	10	10
DNL (LSB)	0.34	0.97	0.5	0.44
INL (LSB)	0.62	1.9	2.2	0.45
Power (nW)	500	97	1900	84
ENOB (bit)	8.72	10.1	8.75	8.95
FOM (fJ/c.-s.)	2.37	2.2	4.4	0.85
Active Area (mm^2)	0.042	0.076	0.026	0.0065

Figure 11.2.6: Performance summary and comparison.

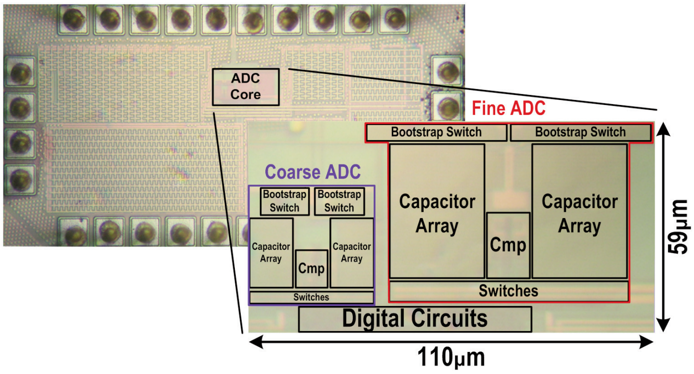


Figure 11.2.7: Chip micrograph.